

LANGDON BAY, United Kingdom

WMO: 037960

Lat: 51.134N

Lon: 1.343E

Elev: 117

StdP: 99.93

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.8	-0.7	-5.4	2.4	-0.1	-4.0	2.7	0.6	18.6	5.8	16.4	7.0	7.6	330	0.728

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	23.9	18.6	22.0	17.5	20.5	16.8	19.2	22.6	18.3	20.8	17.5	19.6	4.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.1	20.6	17.2	12.5	19.4	16.6	12.0	18.6	55.2	22.7	52.1	21.0	49.7	19.6	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.3	13.7	12.3	DB	-3.6	27.8	1.5	2.5	-4.6	29.6	-5.5	31.1	-6.3	32.5	-7.4	34.4	
			WB	-4.3	21.2	1.5	1.1	-5.4	21.9	-6.3	22.6	-7.2	23.1	-8.3	23.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	5.1	5.0	6.6	8.9	11.8	14.2	16.6	16.9	15.0	12.1	8.2	5.8
	DBStd	4.99	2.79	2.80	2.55	2.68	2.56	2.27	2.24	2.02	2.03	2.45	2.85	3.08
	HDD10.0	671	153	141	108	51	10	0	0	0	9	67	132	
	HDD18.3	2868	412	374	364	283	202	126	64	56	103	192	304	388
	CDD10.0	870	0	0	2	18	66	127	205	213	149	76	13	2
	CDD18.3	26	0	0	0	0	0	3	11	11	2	0	0	0
	CDH23.3	81	0	0	0	0	1	10	35	32	3	0	0	0
	CDH26.7	9	0	0	0	0	0	1	5	3	0	0	0	0
(14) Wind	WSAvg	6.4	7.6	7.2	6.8	6.1	6.1	5.5	5.2	5.4	6.0	6.7	7.2	7.4
(15) Precipitation	PrecAvg	705	65	51	40	45	49	51	50	57	57	83	87	71
	PrecMax	980	143	119	103	134	124	117	96	124	105	212	163	122
	PrecMin	541	18	10	6	6	7	14	14	13	5	13	31	16
	PrecStd	105	29	29	22	29	27	30	22	34	29	43	38	32
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.6	15.1	20.1	21.9	24.6	26.6	26.7	23.0	19.0	15.1	12.6
		MCWB	10.0	9.1	11.2	13.7	16.4	18.8	19.9	19.4	18.2	16.1	13.5	11.5
	2%	DB	10.4	10.2	12.7	16.2	19.6	21.4	23.8	23.8	20.2	17.3	13.9	11.7
		MCWB	9.5	8.6	9.5	11.8	15.1	16.9	18.7	18.7	16.9	15.3	12.7	11.0
	5%	DB	9.7	9.4	11.1	14.2	17.6	19.4	22.0	21.5	19.1	16.4	13.1	11.1
		MCWB	9.0	8.2	8.7	10.8	13.9	15.9	17.6	17.7	16.2	14.8	12.0	10.3
	10%	DB	9.1	8.8	9.9	12.6	15.8	17.9	20.2	20.0	18.1	15.6	12.2	10.3
		MCWB	8.3	7.8	8.2	9.9	12.9	15.1	16.9	16.8	15.6	14.2	11.1	9.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	9.9	11.5	14.4	17.4	19.4	20.9	20.6	18.8	16.6	14.0	12.0
		MCDB	10.9	10.5	14.2	18.5	20.8	23.6	25.3	24.9	21.2	18.1	14.5	12.2
	2%	WB	9.9	9.2	10.2	12.4	15.6	17.7	19.3	19.2	17.7	15.9	13.1	11.3
		MCDB	10.1	9.7	11.8	15.4	18.5	20.7	22.8	22.3	19.5	16.8	13.6	11.6
	5%	WB	9.2	8.6	9.3	11.2	14.3	16.4	18.2	18.2	16.9	15.2	12.2	10.6
		MCDB	9.6	9.2	10.5	13.4	16.8	18.5	20.9	20.5	18.3	16.0	12.9	11.0
	10%	WB	8.4	8.0	8.6	10.4	13.2	15.5	17.3	17.5	16.2	14.6	11.2	9.6
		MCDB	9.0	8.6	9.6	12.1	15.2	17.4	19.5	19.4	17.6	15.4	12.0	10.2
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	4.1	5.2	5.8	5.9	5.7	5.8	5.8	5.6	4.8	4.3	3.9
		MCDBR	4.1	4.6	6.7	8.0	8.5	8.1	8.5	7.5	6.3	5.0	4.3	3.8
	5% WB	MCWBR	4.3	3.9	4.6	4.8	5.2	5.0	4.6	4.0	4.1	3.8	3.9	4.0
		MCWBR	4.1	4.2	5.8	7.0	7.7	7.3	7.5	6.4	5.5	4.4	3.9	3.7
(40) Clear-Sky Solar Irradiance	taub		0.328	0.344	0.380	0.398	0.406	0.406	0.404	0.396	0.376	0.361	0.335	0.320
		taud	2.402	2.404	2.304	2.285	2.315	2.335	2.348	2.391	2.434	2.451	2.433	2.391
	Ebn at Noon		688	780	811	840	850	851	846	835	811	750	673	634
		Edh at Noon	62	80	105	120	121	120	117	107	92	76	60	54
(44) All-Sky Solar Radiation	RadAvg		0.81	1.49	2.66	4.28	5.08	5.64	5.38	4.48	3.35	1.90	0.95	0.62
	RadStd		0.07	0.22	0.33	0.38	0.36	0.45	0.45	0.32	0.24	0.18	0.11	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page